

Research Assignment 1: Foundations of Data Analytics & Data Science

1. A database is where we store data. Databases are important because they allow easy access to the information and secure storage. Databases are used in various settings such as a restaurant and Home affairs.

2. A data warehouse can be defined as a central place where information is stored for analytical purposes. It differs from a regular database because it is the umbrella for multiple databases. Companies store data at the data warehouse, to access it and analyse for example, Picer Pay has a data warehouse, that has all the information about how many products were sold per day and how the specials are doing. Therefore, at the end of each day, Data analytics access that information, to help the business grow.

3. A data lake is a database aimed at capturing raw data. Any type of data can be posted or captured. for example, videos, graphs etc. On the other hand, a data warehouse has data that has been processed through the ETL process, therefore ready for analytical processes.

4. A data mart is a mechanism that is used by business ^{users} to access data from the data warehouse. It is not every business user that will access such data. for example, in a company, an accountant will only look at the data that goes in line with his work, to ^{make} ~~affect~~ the business operate successfully. Whereas, a marketer, will look at what convinces buyers to buy or purchase a certain product.

5. Structured data is data stored in a column and includes rows. for example, excel. This data is easy to query with SQL.

Unstructured data, is data that we can consider as "junk". It does not have a structure and it is not organised nor have a consistent format. Therefore, relational Database Management systems can not

score this data. For example: photos, emails etc. Lastly, semistructured data does not necessarily have a definition. It falls in between structured and unstructured data. For example, a music audio file.

6. Data modelling involves creating a visual representation of how data has been stored in a system and the relationship the data has. This process involves describing the characteristics of the data. It is important because it is aimed at meeting the business' needs. The first type is conceptual data models. The models are used during the beginning of a project and will include information about which includes the aspects of the business that will be important to include in a data model. Secondly logical data models provide a detailed explanation of the concepts and relationships of data.

7. Data mining is the use of machine learning and statistical analysis to uncover patterns and other valuable information from large data sets. It is used to analyse both structured and unstructured data. It is mostly used to analyse consumer behaviour or to analyse sales.

Examples: during covid-19 or other universities continue to collect data about their students. They can send a survey or questionnaire that wants to find out how many students travel on a daily basis to come to campus and whether they have access to wifi, when they study. During covid-19 pandemic, a lot of institutions used these methods to find out about their students' backgrounds and where they could possibly improve the learning challenges some students were faced with.

Secondly, data mining can be used in understanding how users find a certain app by means of a feedback. This will help the business users to understand what features of the app people could be categorizing to use or understand.

8. A DBMS is a system that manages the information in a database.

9. The process of gathering data from different sources, is referred to as Extract. Secondly, Transforming involves processing the information, deleting the unnecessary data and storing the relevant one. This is done through querying in SQL. Thirdly, Loading means finally taking out the data or distributing it to the relevant sources.

This is an important step as it allows data analysts to access the data in the data warehouse. It comes with useful insights for the data analysts to work with to meet business needs.

10.

11. Artificial intelligence is intelligence that involves systems that can perform tasks humans can normally perform. Weak AI is for performing a specific task, therefore, whenever it is programmed for, it performs.

It cannot perform any other tasks outside of those. For example:

Siri. General AI, is AI that has been programmed to understand concepts on a larger scale. It involves learning and applying knowledge.

Therefore, it is not limited to specific tasks. For example: a robot that can do different tasks.

12. Machine learning is a AI subset that enables systems to learn and improve from data without additional programming interventions.

* Supervised machine learning is a type of machine learning where the model is trained for a specific task or outcome.

* Unsupervised: It uses machine learning algorithms to analyse and cluster unlabeled data sets. This includes hidden data maybe from semi-structured data.

* Self-supervised machine learning : enables models to train themselves on unlabeled data.

Machine Learning is different from Traditional programming because it works with structured and unstructured data.

Sources :

1. Alex the Analyst (Youtube)
2. AWS website
3. IBM
4. Snowflake
5. SQL for Data Analysts (Advanced Techniques for Transforming Data into Insights)
6. IBM Technology (Youtube)
7. Coursera